

Statement of intent

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PhD position, auction-based support schemes for renewable energy
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I am applying for the doctoral position on auction-based support schemes for renewable energy in Europe. This statement sets out the training I would bring to it and the training I would have to acquire.

Background in energy

My energy training is at Cologne, where the specialisation is taught by the Institute of Energy Economics. I took *Energy Markets and Regulation* with Jun.-Prof. Dr. Oliver Ruhnau, whose chair is in energy market design, and finished it with a grade of 1.3 on a scale where 1.0 is the maximum. The course covered electricity market design, network regulation, and the instruments used to bring renewable generation into a market that was not built for intermittent supply. I then took *Growth, Energy and Climate Change* with PD Dr. Dietmar Lindenberger.

My final seminar is the Panel Data Methods seminar at Cologne, and my paper there is on the EU Emissions Trading System and firm production decisions. The seminar work compared the identification strategies the firm-level ETS literature uses: coarsened exact matching with difference-in-differences in Colmer, Martin, Muûls and Wagner (2025), propensity-score matching on pre-treatment trends in Dechezleprêtre, Nachtigall and Venmans (2023), an instrumental-variables design built on carbon-leakage criteria in Martin, Muûls, de Preux and Wagner (2014), and heterogeneity-robust staggered difference-in-differences in Bremer and Sommer (2025), who apply the Callaway and Sant'Anna estimator to Dutch firms across Phases I to III.

That last paper matters for this application. The estimator Bremer and Sommer apply to European carbon regulation is the estimator I applied to Brazilian administrative data in my thesis, and the reason is the same in both cases: the treatment arrives at different dates with different intensity, and conventional panel estimators mishandle that. The identification problems the seminar catalogued also recur in auction design. Anticipation, because the ETS Directive was adopted in 2003 and the scheme began in 2005. Confounding from the Great Recession. Violations of SUTVA when regulated firms affect unregulated ones through product markets. Free allocation weakening effective treatment intensity, so that the allowance price overstates the carbon cost a firm actually faces.

The gap the seminar left me with is that credible firm-level evidence concentrates on Phases I and II, when allowance prices were low and free allocation generous, and that evidence for Phases III and IV remains scarce.

Proposed contribution

Of the four directions in the call, the two I would want to take up are pre-qualification requirements together with completion bonds, and local content requirements. My reason is specific to the identification problem they present.

Auction design in Europe is not a single policy. Member states adopted auctions at different dates and subsequently revised the rules at different dates again: penalty levels, bid bond

requirements, realisation deadlines, technology-specific against technology-neutral lots, and local content conditions have each been introduced and amended on separate national timetables. This is staggered adoption of a treatment whose intensity varies across adopters, and it is the setting in which conventional panel estimators fail in a way that is easy to miss.

My master's thesis was built for exactly that setting. It estimated the labour-market effects of a national scholarship programme across 558 regions over the period 2005 to 2019, using the heterogeneity-robust staggered difference-in-differences estimator of Callaway and Sant'Anna (2021) with doubly-robust inference, on a panel of roughly five million linked employer-employee records that I constructed. The doubly-robust estimate is a wage effect of +3.04 per cent in low-exposure regions. The two-way fixed effects estimate on the same panel is -0.65 per cent. The sign reverses because two-way fixed effects draws comparisons between later-treated and already-treated units, and those comparisons are invalid once effects vary across adoption cohorts and over time, as Goodman-Bacon (2021) shows. The thesis was graded 1.0.

The design question I would want to answer at CORE follows from this. Auction theory delivers predictions about how pre-qualification and penalty regimes should change bidding behaviour and completion rates. Testing those predictions against the European record requires an estimator that survives staggered adoption with heterogeneous treatment effects, and the diagnostics that establish whether it has survived. That is the part of the project I could do on arrival.

Training I would have to acquire

The call asks for graduate training in microeconomic theory, industrial organization and game theory. My graduate coursework is econometric. My strongest marks are in econometrics, mathematics for economists and energy regulation, and my microeconomics grade is 3.0, which is average on the German scale. I would begin the theoretical component of this project from a lower base than a candidate trained in industrial organization, and the transcript accompanying this application shows that.

Two things qualify it. Mas-Colell, Whinston and Green and Osborne and Rubinstein are the standard the call names, and reaching that standard is a matter of coursework and reading that a first year is designed to accommodate. The empirical craft is the harder thing to add later, and I have it already.

Practicalities

I complete my M.Sc. in Economics at Cologne this month and am available from October 2026, which fits the autumn start. I hold IELTS Academic 8.0, and I speak German and Tunisian Arabic natively and French at B1. I hold EU citizenship, so no work permit is required.

My referees are Prof. Dr. Pia Pinger, who chaired my master's thesis, and Jun.-Prof. Dr. Oliver Ruhnau, who taught and examined me in energy market design. Contact details appear on the CV. The thesis and the seminar paper on the EU ETS are available on request.

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